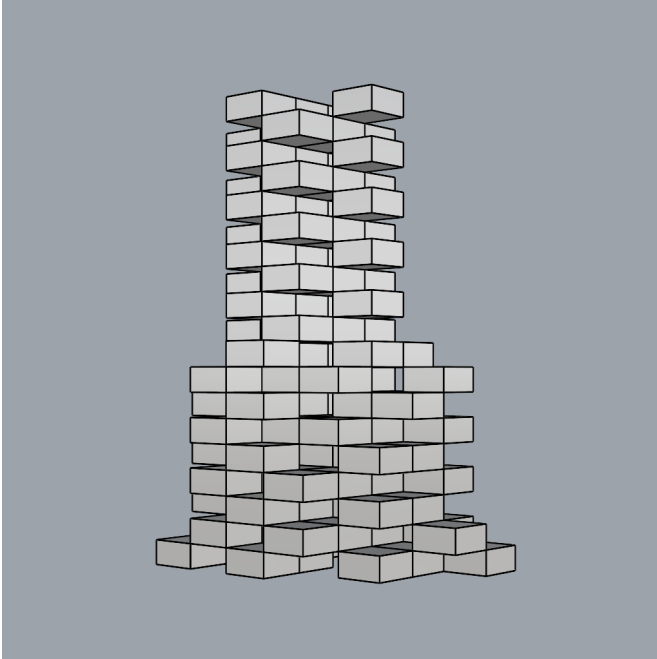


A Framework for Architectural Geometry Synthesis Using Cellular Automata and Conditioned Diffusion Models

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(a) Procedural Voxel Model



(b) Conditioned Diffusion Rendering

Figure 1: Our core contribution: translating a procedurally generated 3D Cellular Automata voxel model (a) into a high-fidelity architectural rendering (b). The ControlNet-conditioned diffusion process preserves the essential geometric structure, including massing, voids, and silhouette, while synthesizing photorealistic materials and lighting.

Abstract

We present a framework for generating 3D architectural geometry and corresponding high-fidelity renderings from procedural rules. Our pipeline integrates a 3D Cellular Automata (CA) engine for voxel massing, a geometry-conditioned diffusion model (SDXL with ControlNet) for structure-aware rendering, and a Large Language Model (LLM) for automated metadata generation. This system links procedural logic with visual design, enabling rapid exploration and the creation of structured datasets. We demonstrate the framework

by generating over 15,000 unique configurations at 70 seconds per model.

CCS Concepts

- **Computing methodologies** → **Image processing**; *Animation*;
- **Applied computing** → *Media arts*.

Keywords

procedural generation, cellular automata, diffusion models, ControlNet, prompt engineering, computational design, generative AI, architecture

ACM Reference Format:

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1 Introduction

A significant challenge in digital design is the gap between abstract, rule-based generation and the production of detailed visual assets. In architecture, this often decouples early-stage massing exploration from final visualization. While recent text-to-image models offer powerful image synthesis capabilities, they lack the fine-grained structural control necessary for architectural applications. We address this gap with a structure-aware framework that couples procedural geometry with conditioned image synthesis. Our contributions include an automated CA→Rendering→Labeling workflow and a scalable data generation loop that programmatically pairs geometry, renderings, and LLM-generated descriptions for downstream learning tasks.

2 System Architecture and Workflow

Our system is implemented as a four-stage pipeline (Figure 2) using Python, with CAD scripting for geometry processing and a node-based diffusion backend for rendering [comfyanonymous 2023].

2.1 Voxel Geometry via 3D Cellular Automata

We evolve a $64 \times 64 \times 64$ voxel grid from a seeded state under a set of ‘birth/survival’ rules (e.g., B56/S567) [Wolfram 1983]. This method produces a diverse set of tower-like geometries with varied porosity and form. We record the rule set, seed, and final voxel count for each model.

2.2 Conditioned Photorealistic Rendering

An isometric image of the voxel massing is used to guide a Stable Diffusion XL (SDXL) model [Podell et al. 2023]. We use ControlNet [Zhang and Agrawala 2023] with a Canny edge detector as a geometric constraint, which forces the renderer to adhere to the voxel model’s silhouette while synthesizing materials and lighting. The process utilizes architecture-specific checkpoints such as ArchitectureRealMix [laizhende 2023].

2.3 Automated Prompt and Metadata Generation

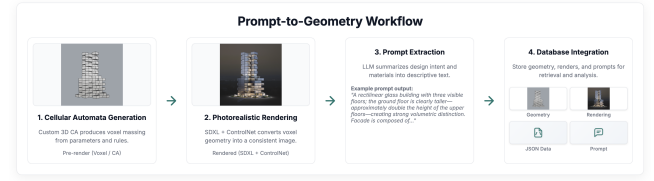
The rendered image and its corresponding CA parameters are passed to an LLM (GPT-4.1) [OpenAI 2025]. The LLM is prompted to provide a detailed architectural description of the image, effectively “reverse-prompting” the design. This process creates high-quality, descriptive text labels that are directly paired with the visual and geometric data.

2.4 Data Curation and Scalability

Each generated record, containing the voxel data, final rendering, CA parameters, and LLM-generated text, is stored in a structured JSON/CSV format. The pipeline is designed for batch execution and can produce over 10,000 unique, labeled variations with deterministic seeding to ensure reproducibility.

3 Results and Evaluation

Our framework was validated by generating a dataset of over 15,000 unique configurations. On a single NVIDIA A40 GPU, the end-to-end generation time averages approximately 70 seconds per model.



(a) The four-stage data pipeline.



(b) A gallery of diverse results.

Figure 2: System architecture (a) and generated results (b).

The primary success of our method is the high fidelity of structure preservation, as shown in the teaser figure. As a preliminary quantitative check, we compare projected voxel silhouettes with Canny edges from the final renders and observe high overlap in visible regions; minor discrepancies occur in occluded areas not captured in the rendered view. The ControlNet-based conditioning correctly translates the abstract voxel massing into a detailed rendering without losing the essential volumetric form. The system is capable of producing a wide array of architectural styles, from tapered monoliths to fragmented stacks. Qualitative review confirms that the LLM-generated descriptions are coherent and accurately reflect the visual output, making them suitable as training labels.

4 Conclusion

We have presented a compact and scalable framework that successfully links procedural generation with geometry-aware diffusion models to produce high-quality architectural imagery. The automated labeling loop creates a valuable flywheel for generating large-scale, structured datasets. Architectural plausibility (scale cues, constructability, and code constraints) is not yet assessed and will be evaluated with domain experts in future work.

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